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B.E. SEMESTER II EXAMINATION (JUNE-2023)

Subject Code: CC102-N

Subject Name: Basic Electrical and Electronics Engineering

Date: 22/06/2023

Time: 10:00 am to 01:00 pm

Total Marks: 70

Instructions:

1. Answer each section in good handwriting in separate answer sheet.
2. All questions are compulsory.
3. Use of scientific calculator is permitted.
4. Indicate clearly the options you attempt, along with its respective question number.

SECTION -I

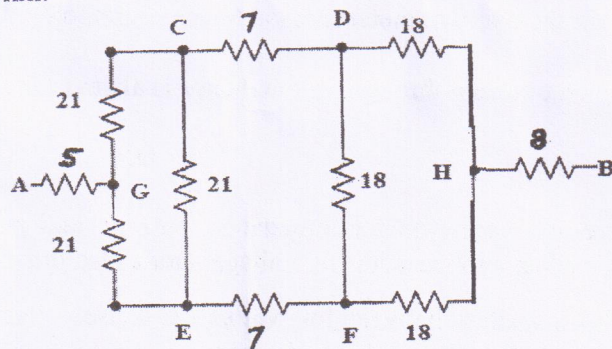
Q-1 [A] Derive expression for delta to star conversion of resistive network. [05]

[B] Explain Ohm's law with its limitations. [05]

[C] Explain KCL and KVL. (Kirchhoff's law) [05]

OR

[C] Find the equivalent resistance between terminal A and B of given network. Consider all resistances are in ohm. [05]



Q-2 [A] Explain the given terms:- 1) Magnetic flux 2) Magnetic flux density 3) Magneto motive force 4) Magnetic field intensity 5) Reluctance. [05]

[B] Explain the difference between electric circuit and magnetic circuit. [05]

OR

Q-2 [A] Derive the expression for equivalent Capacitance of a group of Capacitors when (1) They are Connected In Parallel And (2) They are Connected In Series. [05]

[B] Explain Joule's law of electrical heating. [05]

Q-3 [A] Explain effect of temperature on resistance of: (i) Pure metals (ii) Alloys (iii) insulators, Semiconductors. [05]

[B] Explain Faraday's law of electromagnetic induction. [05]

OR

Q-3 [A] Prove that average power consumption in pure capacitor is zero. [05]

[B] Explain Charging Of Capacitor. [05]

SECTION -II

Q-4 [A] Derive an expression for impedance, current, and power factor for an R-L series circuit when applied with a.c. voltage. Draw also the phasor diagram. [05]

[B] What is coefficient of coupling? Derive expression for the same between two magnetically coupled coils. [05]

[C] Explain how ac sinusoidal emf is generated? Derive its equation of $e = E_m \sin \omega t$. [05]

OR

[C] Explain the R-L-C series resonance phenomena. [05]

Q-5 [A] Explain the two wattmeter method for measurement of 3-phase power. [05]

[B] Prove that current through pure inductor is always lagging by 90° to its voltage. [05]

OR

Q-5 [A] Derive an expression for impedance, current, and power factor for an R-L-C series circuit when applied with a.c. voltage. Draw also the phasor diagram. [05]

[B] Establish relation between line voltage & phase voltage and current relation in 3-phase star connection. Draw phasor diagram. [05]

Q-6 [A] Draw and explain V-I characteristics of P-N junction Diode. [05]

[B] Draw and explain half wave rectifier circuits. [05]

OR

Q-6 [A] Write Short note on Bipolar junction transistor (BJT). [05]

[B] Write short note on Zener Diode. [05]